

Find and Fix: Does the Data Really Show a Cause?

Directions

- Two quantities are **associated** when they rise or fall together. That is a pattern in the data, and nothing more.
- A **cause** is a much stronger claim. Data collected by watching (a survey) can show association; only a fair experiment, where the researcher decides who gets what, can support a cause.
- Watch for a **third factor** that pushes both quantities at once, and for a claim that has the direction backwards.
- Where a student's work is shown, correct it *and* say what went wrong.

1. A seaside town recorded six summer weeks.

Week	1	2	3	4	5	6
Ice creams sold (hundreds)	12	18	25	30	22	15
Beach rescues	1	2	4	5	3	3

Find the mean number of beach rescues per week.

Answer: _____

2. Five seaside towns reported one summer's figures.

Town	Ashby	Byrne	Cardel	Dunmore	Eller
Ice creams sold (hundreds)	20	35	48	27	40
Beach rescues	2	4	7	3	5

Ola wrote: "The town that sells the most ice cream also has the most rescues, so buying ice cream causes people to need rescuing."

- (a) The greatest number of rescues in any town is _____

- (b) Describe what is wrong with Ola's conclusion.

3. The same five towns.

Town	Ashby	Byrne	Cardel	Dunmore	Eller
Ice creams sold (hundreds)	20	35	48	27	40
Beach rescues	2	4	7	3	5

- (a) Cardel had how many more rescues than Ashby? _____

- (b) Describe a third factor that would push *both* rows of the table up at the same time, and say how it affects each row.

4. A survey of 200 students found that 60 of them play a musical instrument, and that 45 of those 60 scored above average on a reading test. Jack was asked what percent of the instrument players scored above average. He wrote:

$$\frac{45}{200} \times 100 = 22.5\%$$

(a) Work out the correct percentage. _____

(b) Describe the mistake Jack made.

5. A survey split students by how many books they own at home and recorded their test scores.

More than 100 books	78	85	82	91	84
Fewer than 100 books	70	75	68	77	70

A newspaper headline read: “Buying books raises test scores.”

(a) The mean score of the more-than-100-books group is _____

(b) Explain whether the survey justifies that headline.

6. The same book survey.

More than 100 books	78	85	82	91	84
Fewer than 100 books	70	75	68	77	70

(a) The mean score of the fewer-than-100-books group is _____

(b) Describe what evidence would be needed before anyone could fairly claim that owning more books *raises* a student's score.

7. Five towns reported their fire services for one year.

Town	Alton	Brent	Chase	Delph	Ely
Firefighters employed	6	10	14	22	30
Fire damage (thousands of dollars)	40	70	95	160	210

Rita wrote: "Towns with more firefighters have more fire damage, so employing firefighters causes damage."

(a) Ely's damage exceeded Alton's by _____ thousand dollars

(b) Describe what is wrong with Rita's reasoning.

8. Two studies were run at the same school.

Study 1. Ten students were split into two groups *at random*. One group followed a fixed sleep schedule for a month; the other carried on as usual. Their end-of-month scores were:

Sleep schedule	82	88	79	91	85
Carried on as usual	74	80	71	83	77

Study 2. Students were asked whether they use a study app. Those who said yes scored higher on average than those who said no.

Mean of the sleep group _____ Mean of the other group _____ Difference _____

Explain which of the two studies could support a claim that one thing *causes* the other, and why the other one cannot.
